



Load balancing in 5G heterogeneous networks based on automatic weight function

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Abstract

Load balancing is a major challenge in heterogeneous networks (HetNets) consisting of 5G and 6G ultra-dense small cells with long-term evaluation advanced (LTE-A) networks. A key factor in achieving efficient load balancing during user mobility is creating appropriate optimisation for handover control parameters (HCP). This paper proposes a coordinated load balancing algorithm for LTE-A/fifth generation (5G) HetNets. The algorithm automatically optimises HCP settings for a given user based on three bounded functions (the signal-to-interference-plus-noise ratio (SINR) of the user equipment (UE), the number of physical resource blocks (PRBs) per UE and the UE's speed) as well as their automatic weight levels. A two-step target cell determination strategy is implemented according to the cell load level and RSRP criteria, ensuring that users are handed over to low-loaded target cells. A new HO procedure that considers the pilot signal power is also proposed, which includes the number of PRBs per UE and the RSRP. Cells with fewer PRBs are prioritised in user association to provide load balance and enhanced throughput. The proposed load balancing algorithm is compared with five other load balancing algorithms selected from the literature. The simulation results reveal that under various mobile speed scenarios, the proposed load balancing scheme enhances network performance in terms of load level, throughput, spectral efficiency and call dropping ratio (CDR).

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1. Introduction

The massive growth in the use of smart devices and applications has substantially increased the need for mobile broadband services with faster data rates and better service quality [1,2]. Next-generation networks must be exceedingly heterogeneous, densely deployed and highly dynamic due to immense data traffic demands of mobile users and the absolute necessity for good communication service quality [3,4]. One of the main challenges of mobility management in 5G

systems and future HetNets is ensuring a fair load distribution between cells [5–7]. Load balancing is a promising approach for effectively handling higher data rates and increased system capacity since it equitably distributes cell loads among neighbouring cells and manages wireless resource allocation across numerous links. Small cells are used to increase capacity in high-demand areas, fill “coverage gaps” in the macro cell layer and reduce load levels in congested cells. In HetNets, employing the traditional scheme of user association based on maximum received power leads to severe load imbalance due to the large differences in power transmission between the macro and small cells. Users are typically connected to macro cells that offer the strongest downlink signal. Due to few associated users, small cells with low transmit power will waste resources. In contrast, UEs associated with a heavily loaded macro cell compete for insufficient network resources, resulting in lower network throughput.

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The disparity between capacity requirements and spectrum scarcity is becoming increasingly clear with rapidly growing mobile traffic demands and the wireless bandwidth bottleneck. Thanks to its wide bandwidth range, mmWave cells are used to increase data rate, enhance mobile broadband and provide reliable connectivity. However, mmWave frequencies suffer from severe free space path loss and atmospheric absorption and are extremely sensitive to abrupt channel changes. Since it is unrealistic to assume that exclusively employing mmWave small cells can ubiquitously provide coverage, it will likely be used in conjunction with LTE.

This study focuses on the optimisation of HCP settings (the HO margin (HOM) and time-to-trigger (TTT)), which are the parameters that control the decision-making process of HO in HetNets for load balancing purposes. Modifying HCPs to suitable values will result in a balanced load distribution among cells and a fair data rate allocation among users. This is based on user movements within a cell's coverage, which will improve the quality of service (QoS) and quality of experience (QoE). Even if the connection quality is inferior, a high HOM setting improves the likelihood that the target cell will be chosen over the serving cell, which may cause the handover of some users to the target cell. Since these UEs are typically at the cell's edge, they may experience worse radio conditions in the target cell. With a low HOM setting, the traffic load will not be offloaded from congested cells. Users will then experience low throughput or even call drops due to insufficient radio resources. A low TTT interval will increase throughput and CDR performance since it reduces the time required for the HO of users in a congested cell. However, this setting can lead to higher HO probability (HOP) and HO ping-pong (HOPP). A high TTT setting hinders the transfer of users from overloaded to underloaded cells, which reduces their performance.

The performance of the proposed load balancing scheme is compared with four algorithms from the literature. In [8], Ghatak et al. suggested a two-stage load balancing mechanism based on two biases to adjust the layer and radio access technology (RAT) selection in multi-tier HetNets of LTE-A macro cells and mmWave small cells. In [9], Prasad et al. introduced an adaptive load balancing algorithm that selects different HOM and load balancing values for each user to balance loads in LTE HetNets. In [10], Aghazadeh et al. proposed a cell selection algorithm based on the number of PRBs, SINR and reference signal received power (RSRP) for load balancing in two-tier HetNets. The algorithm uses estimated signal strength and biased pilot signals from different base stations (BSs). In [11], the user association is formulated as a local optimisation problem in macro BS. Only a limited CSI feedback user association scheme was proposed, developing a low complexity successive offloading scheme. A distance-based load balancing algorithm was used which allows users to associate with the nearest BS when load imbalance is detected in HetNets. Although these algorithms enhance load balancing performance, they are not robust or ideal for selecting appropriate HCP values. The current algorithms are inefficient for several reasons. The main reason is that most of these

algorithms were developed for LTE technology, which has different features and requirements than the 5G technology. All previously mentioned algorithms were designed to function according to a central optimisation model. This means that the optimisation process is based on the performance of the entire network, not on the basis of individual user experience. Several existing algorithms do not optimise all HCP settings (HOM and TTT).

The main contributions of this paper are summarised, as follows:

1. A new load balancing algorithm is developed in HetNets to adaptively predict the HCP settings for each user independently based on the input parameters: SINR, number of PRBs per UE and UE speed.
2. A two-step method is proposed to determine the target cell according to the cell load level and RSRP criteria.
3. A modified HO decision algorithm is created to control the HO process between the serving and target BSs.
4. Several load balancing performance measures are used to compare the performance of the proposed algorithm with five other algorithms from the literature.

The remainder of this paper is organised as follows. Section 2 presents the system model and simulation environment. Section 3 describes the proposed load balancing strategy. Section 4 discusses the simulation results, and Section 5 concludes the paper.

2. System model

The system model consists of 117 5G small cells and 39 LTE-A macro hexagonal cells. Figure 1 presents an illustration of the HetNet simulation environment used in this investigation. Small cells are located at the centre of each macro cell sector. This study applied the LTE-A Pro 3GPP Rel.16 specifications released by 3GPP to represent all system settings, mobility model and network environment. Directional and omnidirectional antennas are present in each sector and small cell, respectively.

Several mobile users are placed in each evolved Node B (eNB) at random locations within cell boundaries to simulate a real LTE-A/5G HetNet environment. The traffic load of each eNB is periodically and randomly modified in each cell. Fifteen users were chosen to evaluate the suggested algorithm's performance and compare with existing algorithms from the literature. In the first simulation cycle, 15 users were generated at random locations in cell #1. As shown in Fig. 1, each mobile user moves in only one direction, parallel to each other, between the solid blue lines. The performance of 15 users is measured in each simulation cycle. In this paper, the performance metrics (such as the serving cell load level, load variance, throughput, spectral efficiency and CDR) are the average measurement values from the 15 users per simulation cycle. The simulation began according to the settings mentioned in Table 1, creating the simulated network environment. The mobility model is then defined. During the simulation, the positions and directions of the UEs are periodically updated and the Euclidean distances to the network's eNBs are

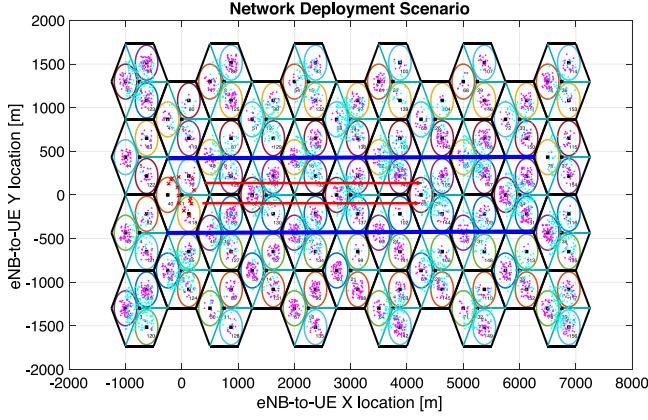


Fig. 1. Simulation environment. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

calculated to form the distance matrix. In multipath scenarios, this distance matrix is used to estimate log-normal shadowing, Rayleigh fading and signal path losses. Each UE's RSRP and SINR are calculated according to the signals received by each carrier. Each eNB updates performance reports such as the load level, throughput, spectral efficiency and CDR during simulation.

Each UE measures the average RSRP and SINR values on the carrier and sends it to the relevant eNB for optimisation. The eNB selects the Modulation and Coding Scheme (MCS) based on the received RSRP and SINR data [12]. The X2 interface-based direct communication between eNBs during HO processes allows the sharing of operational reports and parameter configurations.

3. Proposed load balancing optimisation scheme

This paper proposes a load balancing algorithm that automatically and dynamically estimates appropriate HCP values for each user to achieve load balancing in LTE-A/5G HetNets. Unlike the traditional approach (max-RSRP), the target cell is determined according to the cells' load level and RSRP value. The cells' load level is constantly monitored and a restricted NCL is generated from cells whose load level is below a certain threshold (load level $\leq 50\%$). The cell with the highest RSRP value in NCL is determined as the target cell. This algorithm dynamically estimates HCP settings for each UE based on the UE experience. The algorithm mainly depends on three bounded functions and their auto-weight levels. First, the bounded functions are independently evaluated as a function of the UE's SINR, the number of PRBs per UE and the UE's velocity. Next, the outputs of the three bounded functions are used as inputs in the proposed Automatic Weighting Function (AWF) to estimate the weight of each output bounded function. The final output is then used as an indicator to automatically optimise HCP settings for a particular user.

It is more accurate to formulate the chosen metrics into a single Weight Function based on the variation and weight of each considered factor since they have numerous variations

Table 1
Simulation parameters [15–20].

Parameter	Value	
	LTE-A cell	5G cell
Cell radius (m)	500	200
Number of eNB	61	183
Carrier frequency (GHz)	2.1	28
System bandwidth (MHz)	20	500
Transmit power (dBm)	46	30
Number of PRBs	100	2500
Shadowing standard deviation (dB)	6	7.8
HO preparation time (ms)	50	
HO execution (ms)	40	
SINR threshold (dB)	−10	

and effects. The weight of each bounded function must be considered since each constrained function may have a different effect. Therefore, the Weight Function is calculated as a function of the three bounded functions and their weights. The proposed Weight Function is mathematically expressed in Eq. (1):

$$f_{WF}(\gamma, PRB, v) = \Omega_{\gamma} \times f(\gamma) + \Omega_{PRB} \times f(PR B) + \Omega_v \times f(v), \quad (1)$$

where Ω_{γ} , Ω_{PRB} and Ω_v represent the normalised weight of the UE's SINR, the number of PRBs per UE and the UE's velocity, respectively. The normalised weights can be obtained by

$$\Omega_{\gamma} = \frac{\omega_{\gamma}}{\omega_t}, \quad \Omega_{PRB} = \frac{\omega_{PRB}}{\omega_t}, \quad \Omega_v = \frac{\omega_v}{\omega_t}, \quad (2)$$

$$\omega_t = \omega_{\gamma} + \omega_{PRB} + \omega_v, \quad (3)$$

where ω_{γ} , ω_{PRB} and ω_v are the weights for the UE's SINR, the number of PRBs per UE and the UE's speed, respectively. In the literature, different adaptive HO algorithms have been proposed to adjust HCPs according to Weight Functions [13, 14]. To the best of our knowledge, previous research had assigned different constant weight values. The UE velocity, number of PRBs per UE and the UE's SINR are frequently changed. Constantly assigning static weight settings will lead to erroneous dynamic optimisation. Therefore, a mathematical model, called AWF, is formulated in this paper to dynamically estimate the weight of each bounded function. The AWF model is expressed in Eq. (4):

$$\omega_x = \frac{1 - f(x)}{\sum_{i=1}^F (1 - f(x_i))}, \quad (4)$$

where ω_x represents the weight of the function $f(x)$, which can be $f(\gamma)$, $f(PR B)$ or $f(v)$; therefore, ω_x can be ω_{γ} , ω_{PRB} or ω_v . F is a metric factor representing the total number of parameters considered for optimising HCP values. In this study, F is set to 3 since three parameters (γ , $PR B$, v) are considered. The $f(x_i)$ is a function of x_i , and x_1 , x_2 and x_3 denote γ , $PR B$ and v , respectively.

The first function is SINR $f(\gamma)$: it is the best parameter to represent the received signal strength since it includes noise and interference in its calculations. It assists the BS in making

HO decisions based on the indicator level of received power at the UE, which can be expressed as:

$$f(\gamma) = \frac{\gamma_T - \gamma_S}{\gamma_{max}}, \quad -1 \leq f(\gamma) \leq 1 \quad (5)$$

where γ_T and γ_S represent the SINR on the target and serving eNBs, respectively, while γ_{max} denotes the maximum expected SINR level that can be measured at the UE. The acceptable SINR level ranges from -10 to 30 dB ($\gamma_{max} = 30$ dB).

The second function is the number of PRBs per UE, $f(PRB)$: this function balances the cell load by adjusting HOM values and TTT intervals to produce an HO decision appropriate for a target BS, which can be expressed as:

$$f(PRB) = \frac{PRB_T - PRB_S}{PRB_{max}}, \quad -1 \leq f(PRB) \leq 1 \quad (6)$$

where PRB_T and PRB_S indicate the number of PRBs per UE in the target and serving eNBs, respectively. PRB_{max} represents the maximum number of PRBs per UE.

The third function is the UE speed, $f(v)$: the speed function is an important parameter for adapting HCPs to the UE speed. High mobile speed linearly increases the HO probability. Therefore, the UE speed must be considered when adjusting HCP in order to prevent unnecessary HOs, especially during high movement speeds. This improves throughput and reduces CDR, represented as:

$$f(v) = 2 \log_2 \left(\frac{v_t + v_{max}}{v_{max}} \right) - 1, \quad -1 \leq f(v) \leq 1 \quad (7)$$

where v_t and v_{max} are the UE's velocity and the maximum velocity of the UE, respectively. In this study, it is assumed that the maximum expected velocity of the UE is constant ($v_{max} = 160$ km/h).

Algorithm 1: Handover decision algorithm

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1: Calculate SINR, number of PRBs per UE and UE speed
2: if  $RSRP_t^{pilot} + \Delta HOM > RSRP_s^{pilot}$  then
3:   if Trigger timer  $\geq \Delta TTT$  then
4:     HO decision  $\leftarrow$  True
5:     Send HO request
6:   else HO decision  $\leftarrow$  False
7:     Run Trigger timer
8:   end if
9: else HO decision  $\leftarrow$  false
10:  Reset Trigger timer
11: end if

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The approximated value of $f(\gamma, PRB, v)$ is used to dynamically estimate the appropriate HOM value and TTT duration for each UE independently. The adapted HOM value $\Delta(HOM)$ is expressed as the multiplication of $f(\gamma, PRB, v)$ and the average HOM value ($\overline{HOM}$), or maximum HOM value (HOM_{max}). The $\Delta(HOM)$ is assigned according to the determined conditions based on γ_t , γ_s and threshold SINR values (γ_{th}).

$$\Delta(HOM) = \begin{cases} \overline{HOM} (f(\gamma, PRB, v)), & \text{if } \gamma_{T,S} \leq \gamma_{th} \parallel \gamma_{T,S} \geq \gamma_{th}, \\ \overline{HOM} (1 + \omega_{PRB} f(PRB) + \omega_v f(v)), & \text{if } \gamma_T \leq \gamma_{th} \gamma_S \geq \gamma_{th}, \\ \overline{HOM} (-1 + \omega_{PRB} f(PRB) + \omega_v f(v)), & \text{if } \gamma_S \leq \gamma_{th} \gamma_T \geq \gamma_{th}, \end{cases} \quad (8)$$

$$\overline{HOM} = \frac{HOM_{max} + HOM_{min}}{2}, \quad (9)$$

where HOM_{min} and HOM_{max} indicate the minimum and maximum HOM assigned with the values of 0 dB and 20 dB, respectively. The $\gamma_{T,S}$ represents the SINR level of both the target cell and the serving cell. Similar to HOM, the TTT intervals are dynamically estimated via calculated $f(\gamma, PRB, v)$ values. The update in TTT is represented as ΔT and estimated using the following model:

$$\Delta T = \begin{cases} Z_1 & \text{if } T_{min} < T < T_{max}, \\ Z_2 & \text{if } T = T_{min}, \\ Z_3 & \text{if } T = T_{max}, \end{cases} \quad (10)$$

where Z_1 , Z_2 and Z_3 are represented by Eq. (11), Eq. (12) and Eq. (13), respectively:

$$Z_1 = \begin{cases} T - \rho & \text{if } f_{WF} \leq f_{WF} + Q, \\ T + \rho & \text{if } f_{WF} \geq f_{WF} + Q, \end{cases} \quad (11)$$

$$Z_2 = \begin{cases} T & \text{if } f_{WF} \leq f_{WF} + Q, \\ T + \rho & \text{if } f_{WF} \geq f_{WF} + Q, \end{cases} \quad (12)$$

$$Z_3 = \begin{cases} T - \rho & \text{if } f_{WF} \leq f_{WF} + Q, \\ T & \text{if } f_{WF} \geq f_{WF} + Q, \end{cases} \quad (13)$$

ρ and Q denote the optimisation interval and step level, respectively. The constants ρ and Q control the resolution with which TTT intervals are updated. Higher TTT resolution is attained if these constants are selected as small values. However, very high TTT resolution can increase system latency and computational complexity. For simplicity, the values of ρ and Q are set as 0.04 and 0.1 s. Additionally, T_{min} and T_{max} are determined as 0.0 and 5.12 s, respectively. Algorithm 1 presents the HO decision process.

Here, $RSRP_S^{pilot}$ and $RSRP_T^{pilot}$ are denoted as the $RSRP^{pilot}$ value of the serving and target cell, respectively. To enhance UE and network performance, we propose using $RSRP^{pilot}$ with information on the number of PRBs per UE instead of using RSRP for cell selection. The $RSRP^{pilot}$ is calculated as:

$$RSRP^{pilot} = RSRP_x \frac{N_{PRB,total}}{N_{UE}}, \quad (14)$$

where $N_{PRB,total}$ and N_{UE} denote the total number of PRBs in a cell and the current number of UEs in the cell, respectively.

Thanks to the proposed method, cells with freer PRBs are prioritised in user association, which ensures load balancing. Cells with a greater number of PRBs also provide better throughput and spectral efficiency performance as well as reduced CDR.

4. Results and discussion

This section presents the findings from the simulation study. The proposed algorithm is examined and validated by the simulations conducted using MATLAB 2020b software. The performance results of the suggested algorithm are examined and compared with five other algorithms derived from the literature: Ghatak [8], Prasad [9], Aghazadeh [10], Han [10] and the Distance algorithm. To assess how well the algorithms function in various scenarios, four mobile speed scenarios

Table 2
Simulation results.

		Average load of serving cell		
UE speed (km/h)	40	80	120	160
This paper	0.3791	0.3845	0.3881	0.3655
Ghatak [8]	0.5654	0.5834	0.6012	0.5876
Prasad [9]	0.5236	0.5278	0.5332	0.5238
Aghazadeh [10]	0.4496	0.4642	0.4968	0.4389
Han [10]	0.5611	0.5884	0.5634	0.6021
Dis	0.4730	0.5076	0.5611	0.5501
Load variance				
UE speed (km/h)	40	80	120	160
This paper	0.0415	0.0387	0.0365	0.0355
Ghatak [8]	0.0668	0.0606	0.0624	0.0585
Prasad [9]	0.0652	0.0578	0.0622	0.0594
Aghazadeh [10]	0.0577	0.0549	0.0571	0.0529
Han [10]	0.0651	0.0656	0.0638	0.0622
Dis	0.0511	0.0445	0.0527	0.0385
Average throughput [Mbps]				
UE speed (km/h)	40	80	120	160
This paper	4.6115	4.0827	2.9240	3.4250
Ghatak [8]	2.3994	1.8423	1.0968	1.2951
Prasad [9]	1.8758	1.5887	1.1500	0.9451
Aghazadeh [10]	3.9985	3.0022	1.9272	2.2444
Han [10]	2.2101	1.5163	1.5636	1.4739
Dis	2.9482	3.6172	1.5597	2.4971
Average spectral efficiency [bit/s/Hz]				
UE speed (km/h)	40	80	120	160
This paper	2.4971	2.3710	2.1640	2.1582
Ghatak [8]	2.4740	2.0901	2.0288	1.8133
Prasad [9]	1.7720	1.4551	1.4438	1.1007
Aghazadeh [10]	2.3179	2.1288	1.9663	1.8175
Han [10]	2.4865	2.1624	1.9663	1.8175
Dis	1.4468	1.9477	1.4527	1.6675
Average CDR				
UE speed (km/h)	40	80	120	160
This paper	0.0078	0.0127	0.0175	0.0267
Ghatak [8]	0.0255	0.0273	0.0344	0.0302
Prasad [9]	0.0249	0.0224	0.0537	0.0551
Aghazadeh [10]	0.0123	0.0091	0.0221	0.0246
Han [10]	0.0144	0.0067	0.0230	0.0171
Dis	0.0319	0.0317	0.0681	0.0431

are considered: 40, 80, 120 and 160 km/h. All presented results are the average values of 15 UEs considered in the measurements. Each UE performance is separately assessed for each simulation cycle (40 ms). Table 2 provides the results of the evaluated performance metrics for all proposed and compared algorithms.

Fig. 2 demonstrates the average load level of serving cells for both the proposed and compared algorithms versus the simulation cycle for the UE speed of 80 km/h. In this study, the load level is described as the ratio between the number of users in a cell and the theoretical capacity of the cell at any given time. The results indicate that the proposed algorithm provides the best performance in terms of load balancing at selected mobile speeds. The serving cell's load level of the proposed algorithm varies between 10% and 55%. This proves that the proposed algorithm is successful in offloading user traffic from congested cells to low-loaded cells. Table 2 shows that the proposed algorithm maintains its superior performance compared to the other algorithms for the investigated metrics during different mobile speed scenarios.

Fig. 3 displays the system performance for different load balancing mechanisms in terms of load variance across Het-Nets. The proposed algorithm minimises load variation by dynamically estimating HCP settings for each UE based on the UE's experience. Table 2 provides the average load variances for the proposed solution as well as benchmark solutions.

Fig. 4 displays the average throughput of each distinct mobile speed scenario for all measured users and throughout all simulation cycles. The proposed algorithm achieved the highest average throughput in all mobile speed scenarios. There are two reasons for the proposed algorithm's enhanced performance: (i) the inclusion of SINR and number of PRBs in HCP optimisation, and (ii) the inclusion of resource availability in the HO procedure. The results demonstrate that the proposed algorithm attains performance gains of approximately 15%, 12%, 51% and 37%, respectively, as compared to the algorithms that provide second-best average throughput for various mobile speed scenarios. This is a noteworthy accomplishment.

Fig. 5 presents the average CDR vs simulation cycles for all observed UEs in various mobile speed settings. When

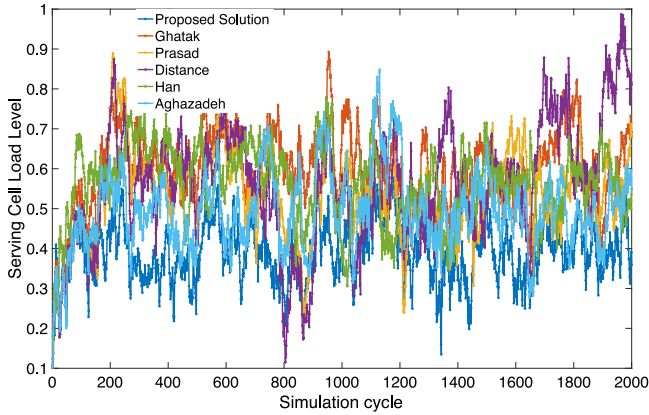


Fig. 2. Average load level of the serving cell at the mobile speed of 80 km/h.

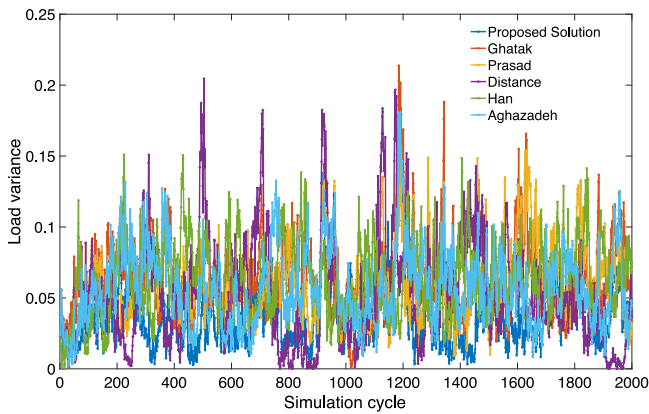


Fig. 3. Load variance at the mobile speed of 80 km/h.

compared to Ghatak, Prasad and the Distance algorithms, the results demonstrate that the proposed scheme offers a discernible reduction gain in CDR. Compared to Aghazadeh and Han, the proposed algorithm performs better at mobile speeds of 40 and 120 km/h, while Aghazadeh and Han perform better at mobile speeds of 80 and 160 km/h. This proves that the proposed algorithm is successful in reducing the call drops that may occur due to poor connection quality and insufficient network resources.

Fig. 6 highlights the average spectral efficiency for different mobile speeds. The results demonstrate that the proposed algorithm significantly enhances spectral efficiency as compared to the other algorithms, especially in high mobile speed scenarios. The proposed algorithm achieved performance gains in spectral efficiency of approximately 0.9%, 13%, 13% and 18%, respectively, throughout various mobile speed scenarios. The fluctuations in signal quality depend on the increase in mobile speed, causing a gradual decrease in spectral efficiency.

5. Conclusion

This paper presented a dynamically coordinated load balancing algorithm for HetNets. The proposed algorithm estimates the HCP settings for each UE independently according

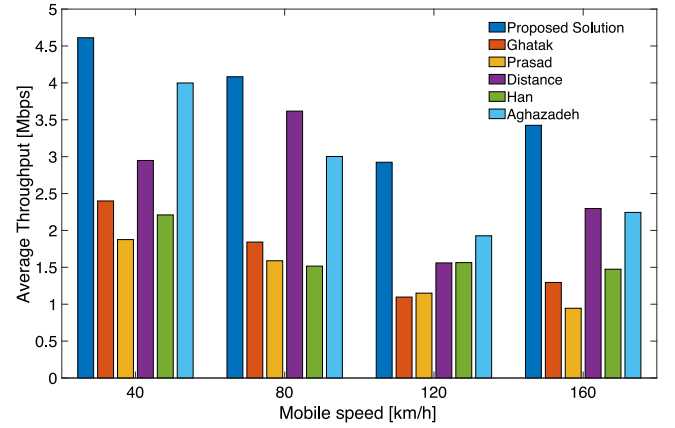


Fig. 4. Average throughput of various schemes according to mobile speed conditions.

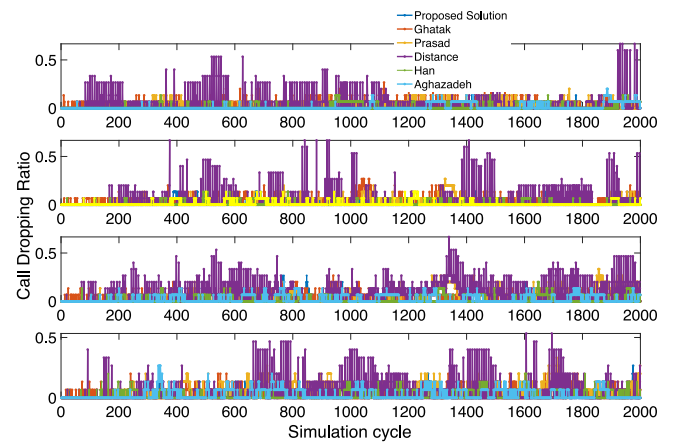


Fig. 5. Average CDR of various schemes in terms of mobile speed scenario.

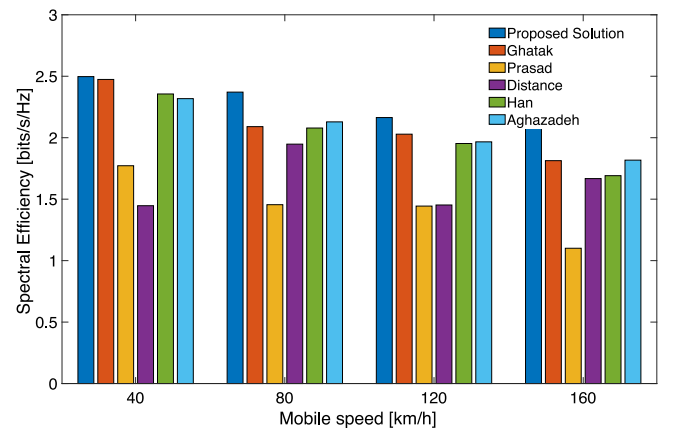


Fig. 6. Average spectral efficiency of various schemes in terms of mobile speed scenarios.

to the Weight Function, based on three bounded functions, and the weight of each bounded function. The bounded functions are considered as the functions of UE SINR, the number of PRBs per UE and the UE's velocity. The weight of each bounded function is calculated using a proposed dynamic mathematical function based on the output of three bounded functions. The proposed scheme also provides flexibility to the

serving network by individually modifying the HCP values of each UE. It adopts a two-step strategy of target cell determination based on the cells' load level and RSRP values. This study further implemented a new HO procedure based on the pilot signal power which considers the RSRP and the number of PRBs per UE. Thereby, cells with more PRBs are prioritised in user association. To evaluate the performance of the proposed algorithm, it was compared with various load balancing algorithms derived from the literature. The simulation results indicate that the proposed scheme performs better in terms of load level, throughput, spectral efficiency and CDR. The findings suggest that coordination strategy and input parameters are crucial for designing effective load balancing models for dense urban HetNets. This will enhance the overall network performance as users move between cells. The extension of this study will consider further assumptions such as system setup scenarios, additional key performance indicators and higher mobile speeds.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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